

Realistic Modelling and Optimization of DTLS Handshakes in Constrained LPWAN Environments

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Abstract—With the growing popularity of Low Power Wide Area Networks (LPWANs) in large-scale Internet of Things (IoT) deployments, ensuring secure yet efficient communication presents a significant challenge. Datagram Transport Layer Security (DTLS) is designed to provide end-to-end security. However, there is a substantial communication overhead associated with DTLS that makes it impractical for resource-constrained environments, such as Low Power Wide Area Networks (LPWANs). This paper presents a simulation-based evaluation of the DTLS handshake performance over two major LPWAN technologies—LoRaWAN and mioty—incorporating realistic network conditions, including packet loss, duty cycle limitations, and message fragmentation. Unlike prior studies based on idealized airtime models, our approach provides actionable insights into real-world performance. We also utilize the Static Context Header Compression (SCHC) mechanism to reduce protocol overhead, achieving around 60% reductions in both handshake duration and message size. Furthermore, we compare the performance of DTLS handshake variants, analyzing trade-offs between certificate-based and Pre-Shared Key (PSK) authentication in constrained environments. Our research is highly relevant for the development of secure and efficient IoT communication protocols by contributing to the ongoing optimization efforts.

Index Terms—DTLS, LPWAN, LoRaWAN, mioty, SCHC, IoT

I. INTRODUCTION

The Internet of Things (IoT) is driving new applications, reducing costs, and increasing efficiency in sectors like smart grids, healthcare, industrial automation, and many more. As IoT continues to grow, with an expected 29.7 billion connected devices by 2027, there is an increasing need for secure and reliable wireless communication technologies [1]. In response to the connectivity challenges presented by IoT, particularly in massive machine-type communications constrained by energy and bandwidth, Low Power Wide Area Networks (LPWAN) have become a prominent solution [2].

However, inherent security mechanisms in LPWANs are often lightweight due to strict energy and bandwidth constraints [3]. Most LPWAN technologies lack a comprehensive end-to-end security framework that spans from the device to the application server, making them vulnerable to eavesdropping, replay attacks, and unauthorized access [4]. This highlights the need for end-to-end security solutions beyond LoRaWAN's native mechanisms.

In this context, Transport Layer Security (TLS) and Datagram Transport Layer Security (DTLS) stand out as potential solutions owing to their established technical maturity,

comprehensive validation through extensive research, and significant adoption across various industries. They enable secure end-to-end communication over untrusted networks [5]. While TLS relies on reliable transport like TCP, DTLS is based on UDP and specifically designed for datagram-based communication, making it more suitable for constrained and lossy networks [6]. DTLS eliminates the overhead of maintaining a persistent connection and integrates mechanisms for handling packet loss and reordering into its handshake protocol [6], [7]. DTLS (Datagram Transport Layer Security) offers security level comparable to TLS (Transport Layer Security), providing robust encryption and authentication while being adaptable to unreliable network conditions [8].

However, the adaptation of DTLS for LPWANs remains challenging due to its multi-flight handshake, large message sizes (especially for certificates), and the need for reliable message delivery. These factors result in excessive airtime, high energy consumption, and increased handshake delays [9]. Prior research has primarily evaluated DTLS performance using analytical airtime models or under idealized conditions, often overlooking the effects of real-world constraints.

To address this gap, we present an event-based simulation model for the evaluation of DTLS 1.2 handshake performance in LPWAN conditions. Building on the framework from [10], our enhanced model integrates realistic radio characteristics—including variable packet loss, transmission delays, and regulatory duty cycles. We also evaluate the role of Static Context Header Compression (SCHC), a compression scheme designed specifically for constrained environments, which offers improved efficiency over traditional IPv6 over Low-Power Wireless Personal Area Networks (6LoWPAN) or Robust Header Compression (RoHC) schemes [11]. The main contributions of this paper are:

- A simplified packet loss-aware modelling approach for DTLS over LPWAN, capturing realistic constraints, has been proposed.
- Quantified impact of SCHC compression on DTLS handshakes in LoRaWAN and mioty environments.
- Comparative analysis of Preshared Key (PSK) versus certificate-based DTLS handshakes, highlighting trade-offs under strict duty cycles.
- Practical design recommendations for secure and efficient LPWAN deployments, including handshake variant selection and compression strategies.

The rest of this paper is organized into the following sections: Section II provides a summary of related work. The simulation framework adopted for this study is described in Section III, followed by the results and key takeaways in Section IV. Finally, conclusions and plans for future work are presented in Section V.

II. RELATED WORKS

The challenge of ensuring end-to-end secure communication in resource-constrained IoT environments such as LPWANs has led to research on adapting existing protocols like DTLS. While DTLS is well-established for internet-scale security, its application in LPWAN contexts poses unique performance and feasibility challenges. This section discusses previous work relevant to LPWAN security, DTLS performance evaluations, and protocol optimizations, while highlighting how our contribution addresses the limitations of existing approaches.

A. Security Challenges and Solutions in LPWANs

LPWAN technologies are known to be vulnerable to various attacks, including Denial of Service, Man-in-the-Middle, and physical tampering, due to their constrained architecture and open wireless medium [12], [13]. These vulnerabilities often arise from shared assumptions across LPWAN stacks, such as a lack of mutual authentication or lightweight encryption. Although research has focused primarily on the security of device-to-gateway communication, end-to-end security threats involving malicious gateways or compromised servers or back-end systems remain underexplored [12]. For example, LoRaWAN employs two layers of security: one between the end devices and the network server to ensure device authenticity, and another between the end devices and the application server to protect data confidentiality [14]. Although payload confidentiality is ensured through application layer encryption, the message integrity verification is performed at the network server, meaning the application server cannot verify the authenticity or integrity of the data it receives. Moreover, since the connection between the network server (typically operated by the LPWAN provider) and the application server (managed by the application owner) often traverses public networks, messages in transit may be susceptible to tampering or redirection unless additional safeguards are in place [15]. In this context, DTLS offers a viable solution that ensures true end-to-end encryption, authentication, and data integrity.

B. Performance Evaluation of DTLS in LPWAN

DTLS handshakes are inherently heavy due to multiple round-trips and large certificate-based messages. In LPWAN scenarios, characterized by low data rates, small payload capacities, and duty cycle restrictions, it leads to significant performance bottlenecks [16]. Prior research has mostly relied on mathematical airtime models to assess the feasibility of DTLS or TLS handshakes in LoRaWAN. For instance, [14] and [17] offer the estimates under ideal transmission conditions. However, such models lack the fidelity to capture practical

network dynamics like packet loss or retransmission overhead due to unreliable wireless channels [18].

The study in [14] illustrates the impact of spreading factors on scalability, showing that at Spreading Factor (SF) 12 only 147 TLS handshakes are feasible in two days per cell, compared to thousands at SF7. Furthermore, it notes that cipher suite selection greatly affects handshake size and latency, with RSA and large ECC curves causing more fragmentation. Similarly, [10] highlights that certificate-based DTLS handshakes are particularly sensitive to packet error rates, confirming the need for efficient authentication mechanisms.

In [17], the authors compare DTLS to Ephemeral Diffie-Hellman Over COSE (EDHOC) protocol and find that DTLS handshakes can take 50–80 seconds at SF12, while EDHOC completes in around 7 seconds. However, EDHOC's performance benefits come from pre-provisioned credentials, reducing flexibility. While these findings emphasize the limitations of DTLS, they do not address optimization possibilities within DTLS itself. Our work builds on this gap by simulating DTLS handshakes with realistic channel conditions, evaluating SCHC compression, and exploring practical variants using PSK-based authentication.

C. Optimization of DTLS

1) *Standardization and Protocol Refinements:* Efforts to make DTLS more suitable for constrained environments are ongoing. The Internet Engineering Task Force (IETF) has proposed adjustments to reduce handshake size and message fragmentation, including recommendations in [19] and profiles for DTLS 1.3 tailored for IoT in [9]. These initiatives suggest optimizations such as improved retransmission timers and minimal state management to accommodate LPWAN conditions.

Additionally, lightweight DTLS implementations have been proposed to reduce cryptographic and memory overhead [20]. While these frameworks are promising, they often need to be combined with complementary techniques, such as compression or stateless reassembly, for practical LPWAN deployments.

2) *Compression Schemes:* Among protocol optimizations, SCHC has gained significant attention for reducing overhead in LPWANs. Unlike stateless compression (e.g., 6LoWPAN) or complex stateful mechanisms (e.g., RoHC), SCHC is designed for predictability and minimal synchronization, fitting LPWAN requirements well [11]. A simulation-based comparison implemented in NS-3 further confirms SCHC's efficiency over 6LoWPAN in LPWAN contexts, particularly for predetermined and fixed data flows [21].

The work in [22] shows that SCHC can reduce header sizes in Constrained Application Protocol (CoAP), DTLS, and Object Security for Constrained RESTful Environments (OSCORE) stacks, leading to notable latency improvements. In smart metering applications, SCHC has reduced handshake duration by up to 40% and energy consumption by similar margins [23]. These results support the relevance of compression in constrained networks, but have not been extensively evaluated

in conjunction with DTLS variants under dynamic loss scenarios, an aspect our study directly addresses.

3) *Industrial Solutions*: Industry contributions have also explored DTLS deployment in LPWAN. A patent by PHYSEC proposes an adaptation layer to mitigate fragmentation and timing issues in LoRaWAN [24], and Acklio reports traffic reduction up to 64% using SCHC compression in NB-IoT networks [25]. While these implementations are valuable, they lack detailed performance analysis across different LPWAN technologies or handshake types.

In contrast, our work evaluates the performance of the DTLS handshake in both LoRaWAN SF12 and mioty under realistic error conditions. With the addition of SCHC to our simulation model and our comparison between PSK and certificate-based handshakes, we bridge the gap between theoretical models and practical limitations, thus achieving a better understanding of the optimization potential for DTLS usage in LPWAN deployments.

III. SIMULATION FRAMEWORK

This research is built upon the simulation framework developed in [10] to assess DTLS performance under challenging conditions, such as high error rates and slow data transmission, utilizing an event-driven model of a client-server connection and a DTLS state machine. We extend this original methodology to incorporate LPWAN-specific characteristics, focusing on two key technologies: LoRaWAN and mioty. Our enhanced model integrates critical LPWAN channel features, including data rate constraints, maximum payload size restrictions, non-airtime latency, and duty cycle regulations. By adapting the simulation to reflect these realistic characteristics, we enable a comprehensive analysis of DTLS handshake efficiency and overall performance in LPWAN environments. The following sections detail these key adaptations, outlining our approach.

Specifically, we model a DTLS communication scenario involving a DTLS server and client operating over a shared medium, subject to duty cycle constraints of 0.1% and 0.01%, respectively. These values are aligned with regulatory transmission limits imposed in the EU 868 MHz ISM band, commonly used in LPWAN deployments [14]. These duty cycle restrictions impact how transmissions are scheduled and managed.

The DTLS handshake operates through a series of grouped message exchanges called 'flights,' each flight requires acknowledgment before the process can advance to the next stage. This structured approach ensures message reliability in networks susceptible to packet loss and high latency. Loss detection and retransmission are managed by treating the reception of any message from the subsequent flight as an implicit acknowledgment of the previous flight's successful transmission. If the transmitter does not receive an acknowledgment within a specified timeout period after repeated attempts, the flight is retransmitted. Special handling is needed for the final flight, as there is no subsequent flights to acknowledge it. In this case, retransmission is triggered upon

re-reception of the second-to-last flight, ensuring reliability in message delivery [10]. Figure 1 illustrates the layered architecture, showing how headers and payloads traverse various layers, including communication layers and dedicated layers for compression and adaptation.

In order to address the specific needs of LPWAN environments, this model integrates SCHC with both Compression/Decompression (C/D) and Fragmentation/Reassembly (F/R) layers. The C/D layer minimizes protocol header overhead to conserve bandwidth, while the F/R layer ensures reliable delivery by adapting larger packets to the small payload limits of LPWAN technologies. These components work together to optimize data transmission by minimizing header sizes and enabling efficient packet fragmentation [23]. SCHC is a generic framework that can be applied across protocol layers, enabling flexible and efficient compression tailored to different networking scenarios. A dedicated SCHC-based DTLS header compression layer is used to effectively minimize the header size as described in [22]. An additional round of SCHC compression is applied to the IP and UDP headers, reducing their combined size from 48 bytes to only 2 bytes. These 2 bytes represent the SCHC Rule ID and the compression residue, as detailed in [22] and [26]. This multi-step compression strategy significantly enhances data transmission efficiency, making it suitable for LPWAN applications. The packet is then divided into fragments to comply with LPWAN's maximum transmission unit (MTU) requirements, such as the 51-byte limit for LoRaWAN SF 12. Each fragment is assigned a minimal 1-byte SCHC F/R header to facilitate reassembly at the destination [26]. Finally, a physical layer-specific header is appended, such as an 11-byte header for mioty, ensuring protocol compatibility.

We modeled the unique characteristics of LPWAN, including data rate constraints, non-airtime latency, and duty cycle limitations, to accurately reflect real-world transmission conditions. The probability of message loss was estimated based on the bit error rate (BER), which varies due to environmental factors, self-interference, and the low-power nature of LPWAN transmissions. To model typical conditions observed in LPWAN environments, we simulated loss rate values up to 50%, which are reported by studies such as [18] and [27] for distances up to 6 km. The impact of fragmentation and compression was assessed by analyzing the total data exchanged between the client and the server and the total completion time of the handshake. For each loss rate, the mean handshake time was calculated over 1000 iterations, along with the standard deviation and a 95% confidence interval to quantify performance variability.

IV. RESULTS AND DISCUSSION

This section evaluates the simulation results, focusing on how DTLS 1.2 handshake performance in LPWAN is influenced by SCHC compression, handshake variants, and varying radio conditions. Our analysis is structured along three key axes: (1) total handshake message sizes and durations, (2) performance differences between PSK-based and certificate-based DTLS

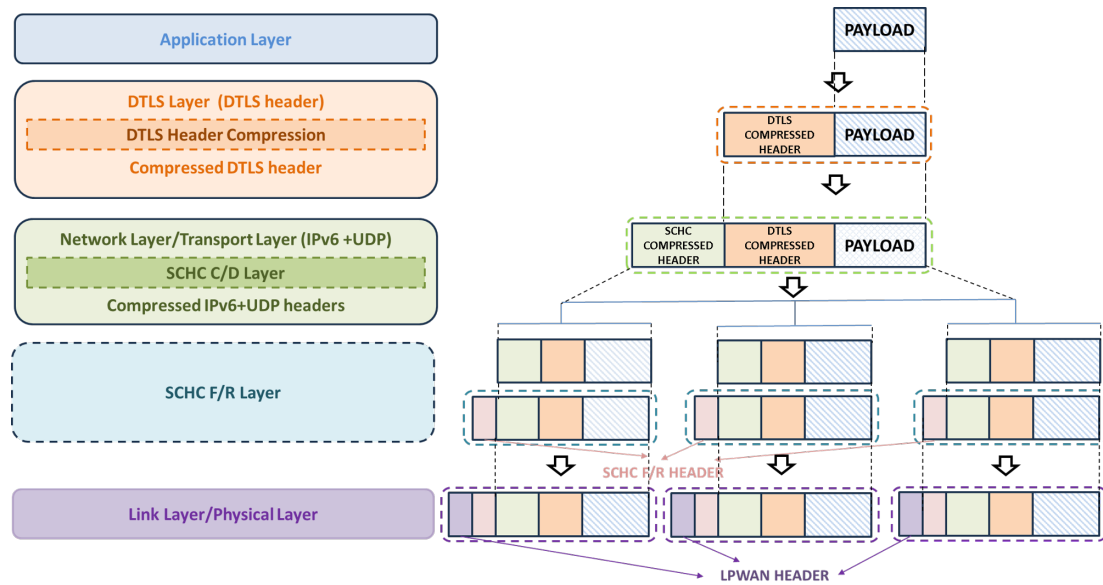


Fig. 1: Protocol Stack Representation and Dataflow

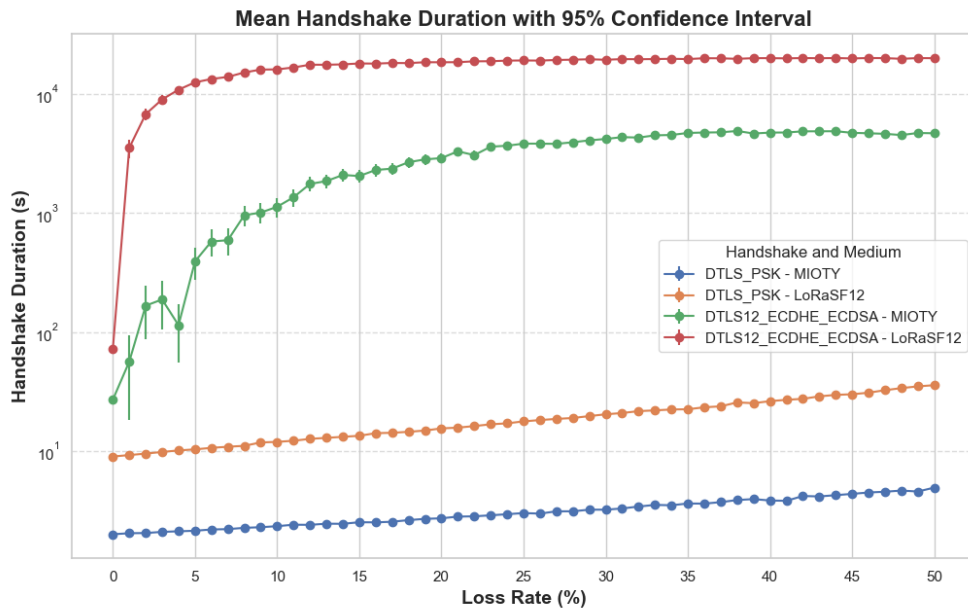


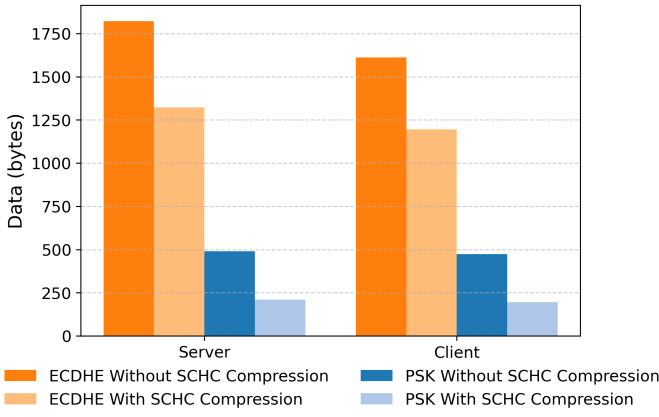
Fig. 2: Variation of DTLS handshake duration with packet loss rate

handshakes, and (3) the impact of increasing packet loss rates. Figures 2, 3, and 4 provide empirical visualizations of these results for LoRaWAN SF12 and mioty networks.

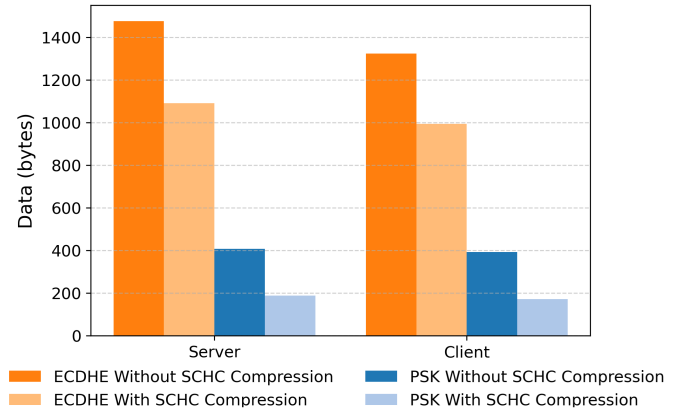
The simulations demonstrate that applying SCHC yields measurable performance benefits. On average, SCHC reduces handshake message sizes by approximately 58%, depending on the handshake variant and communication medium, as shown in Figure 3(a) and 4(a). This compression results in a substantial improvement in the duration of handshakes, with time reductions averaging around 64%, as illustrated in Figure

3(b) and 4(b). These effects are especially pronounced under constrained conditions such as LoRaWAN SF12, where airtime is tightly regulated. This affirms SCHC's practical value in mitigating protocol overhead and improving channel utilization in constrained networks.

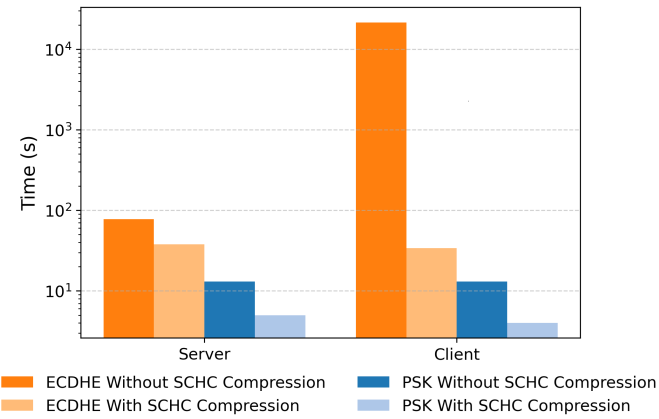
An interesting observation can be made regarding the handshake duration for the DTLS 1.2 ECDHE-ECDSA configuration over LoRaWAN SF12, as shown in Figure 3(b). The client-side handshake time is considerably extended, largely due to the strict 1% duty cycle regulation, which limits the



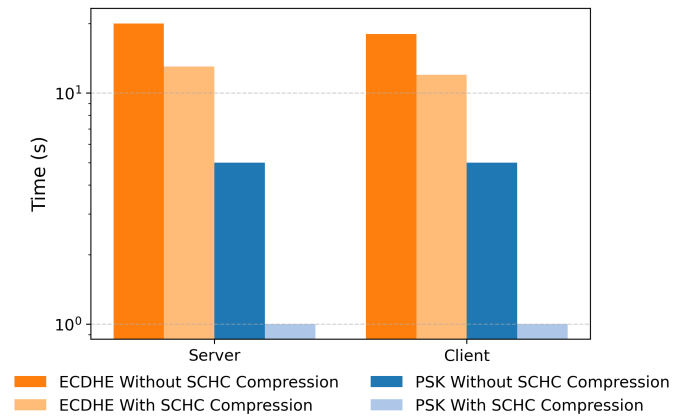
(a) Handshake Data Size (bytes) in LoRaWAN SF12



(a) Handshake Data Size (bytes) in mioty



(b) Handshake Duration (s) in LoRaWAN SF12



(b) Handshake Duration (s) in mioty

Fig. 3: LoRaWAN SF12, Bit Rate: 250 bps, Max. Payload: 51 bytes, Loss rate:0%

Fig. 4: mioty, Bit Rate: 512 bps, Max. Payload: 250 bytes, Loss rate:0%

number of transmittable fragments per unit time. Certificate messages such as *ServerCertificate* and *ClientCertificate* are especially affected, since they require extensive fragmentation. While SCHC mitigates this to some extent by reducing packet sizes, it cannot fully eliminate the overhead imposed by such large payloads under constrained airtime budgets.

In contrast, PSK-based DTLS handshakes perform significantly better under identical conditions. As illustrated in Figures 3 and 4, they exhibit reduced message sizes and completion times due to their simplified exchange structure and lower fragmentation overhead. This makes PSK a practical choice for latency- and energy-sensitive LPWAN applications. However, these benefits come at the cost of limited scalability and inflexible key distribution, making PSK less suitable for dynamic or large-scale networks [9]. Beyond scalability, PSKs also increases the risk of vulnerabilities if individual keys are compromised and complicates revocation in dynamic environments [28]. Despite its overhead, certificate-based authentication remains essential for applications requiring scalable trust models and secure provisioning of new nodes, such as in industrial automation or public infrastructure [9],

[20]. These findings emphasize the importance of selecting handshake mechanisms based on performance metrics, the application's security architecture and lifecycle requirements. Rather than avoiding certificate-based handshakes, future optimizations should focus on reducing their overhead through lightweight credential formats and improved session management.

Figure 2 further illustrates the impact of increasing packet loss rates on handshake duration. mioty exhibits greater resilience to loss, with a gradual performance degradation even under high error rates. This can be attributed to mioty's telegram splitting technology, which reduces susceptibility to interference and supports better fragmentation management [18]. Conversely, LoRaWAN SF12 experiences sharp increases in handshake time beyond a 10% loss rate, with certificate-based handshakes exceeding 10^4 seconds due to repeated retransmissions and prolonged reassembly delays under duty cycle restrictions.

Interestingly, DTLS PSK handshakes exhibit similar robustness across both LPWAN technologies, confirming that they are less sensitive to both fragmentation and packet loss. These findings collectively demonstrate that while SCHC

compression substantially improves protocol efficiency, it is not sufficient on its own to overcome all challenges posed by large message structures. Hence, there is a critical need to pursue complementary optimizations in handshake design and message encoding for use cases demanding stronger authentication guarantees.

V. CONCLUSION AND FUTURE WORK

This paper presents a detailed performance evaluation of DTLS handshakes over LPWAN technologies, focusing on LoRaWAN SF12 and mioty. By integrating SCHC compression into a simulation setup, which emulates the LPWAN-specific constraints, including packet loss, limited MTUs, and regulatory duty cycle limits, we provide a realistic assessment of DTLS feasibility in these environments. Our findings show that SCHC effectively reduces handshake message size by approximately 25–55% and can shorten handshake duration by 30–90%, depending on the protocol configuration and network conditions. These improvements confirm SCHC's role as a practical compression solution for LPWAN deployments.

Among the handshake variants, PSK-based DTLS consistently outperforms certificate-based alternatives in terms of efficiency and stability. However, this improvement comes with trade-offs in security and scalability. Certificate-based handshakes remain challenging under LoRaWAN SF12 conditions, even with SCHC compression, due to their size and retransmission sensitivity. These results highlight the necessity of optimizing handshake structures and selecting authentication methods appropriate to the network's capabilities and application requirements.

An important next step would be validating our findings through hardware implementation and testing. Additionally, it should be noted that while SCHC reduces communication overhead, it can also increase memory usage and processing complexity on constrained devices. Recent research like [29] proposes optimizations such as Layered SCHC (LSCHC), in which contexts are stored separately on each protocol layer. This is an interesting direction for future research.

Future work will also extend our current simulator to explore additional protocol optimizations. In particular, we plan to evaluate Compact TLS (cTLS) [30], a reduced version of TLS 1.3 designed for constrained environments, under similar simulation settings. Moreover, we will assess advanced certificate handling strategies—such as fixed-length encodings, session resumption, Connection IDs, TLS Cached Information (RFC7924), client certificate URLs (RFC6066), raw public keys, and CBOR or compressed certificates (RFC8879)—as recommended in [9].

Another critical direction involves improving DTLS retransmission and backoff strategies. Current methods often rely on exponential backoff and simplistic timeout mechanisms that fail under LPWAN-specific conditions. For instance, the use of fixed retransmission timers based on round-trip time, as recommended in [9], may not be effective when transmissions are delayed due to fragmentation and duty cycles. Furthermore, as shown in [10], message loss in LPWANs is

typically due to corruption rather than congestion, rendering standard TCP-like backoff heuristics suboptimal. We propose to investigate non-exponential, adaptive backoff methods tailored for LPWANs to reduce unnecessary latency while ensuring handshake reliability.

By continuing to evaluate protocol adaptations within realistic LPWAN constraints, we aim to support the broader goal of achieving scalable, secure, and efficient end-to-end communication for the Internet of Things.

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